



Advanced Time Series Analysis and Forecasting

Chapter 12: Machine learning and deep learning for time series



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Acronyms used in this material (1/3)

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ ARX: AutoRegressive model with eXogenous variables
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BPTT: BackPropagation Through Time
- ▣ CDF: Cumulative Distribution Function
- ▣ CPU: Central Processing Unit
- ▣ CRPS: Continuous Ranked Probability Score
- ▣ CUDA: Compute Unified Device Architecture
- ▣ CV: Cross-Validation
- ▣ DGP: Data-Generating Process
- ▣ DM: Diebold–Mariano (test)
- ▣ ENTSO-E: European Network of Transmission System Operators for Electricity
- ▣ ERM: Empirical Risk Minimisation
- ▣ ES-RNN: Exponential Smoothing – Recurrent Neural Network (hybrid)
- ▣ ETS: Error, Trend, Seasonality (exponential smoothing state space models)
- ▣ ETT: Electricity Transformer Temperature (data set)
- ▣ EU: European Union
- ▣ FFT: Fast Fourier Transform
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GRN: Gated Residual Network
- ▣ GRU: Gated Recurrent Unit
- ▣ GW: Giacomini–White (test of conditional predictive ability)

Acronyms used in this material (2/3)

- HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- HAR: Heterogeneous AutoRegressive (model)
- HGB: Histogram Gradient Boosting
- HICP: Harmonised Index of Consumer Prices
- HLN: Harvey–Leybourne–Newbold (small-sample correction of the DM test)
- KKT: Karush–Kuhn–Tucker (conditions)
- LLM: Large Language Model
- LOOCV: Leave-One-Out Cross-Validation
- LSTM: Long Short-Term Memory (network)
- LTSF: Long-Term Series Forecasting
- MA: Moving Average
- MAE: Mean Absolute Error
- MAPAE: Mean Absolute Predictive Accuracy Error
- MASE: Mean Absolute Scaled Error
- MCS: Model Confidence Set
- MIMO: Multiple-Input Multiple-Output (forecast strategy)
- ML: Machine Learning
- MLP: MultiLayer Perceptron
- MPAE: Mean Predictive Accuracy Error
- MSE: Mean Squared Error
- MT: Malta (country code)
- MW: MegaWatt
- NL: NonLinear (shrinkage)
- NVIDIA: NVIDIA Corporation (graphics processors)
- OLS: Ordinary Least Squares
- OOS: Out-Of-Sample

Acronyms used in this material (3/3)

- OWA: Overall Weighted Average (of sMAPE and MASE relative to Naive 2)
- QLIKE: Quasi-LIKelihood loss
- QRF: Quantile Regression Forest
- RF: Risk-Free rate (Fama–French data) / Random Forest (machine learning)
- RMSE: Root Mean Squared Error
- RNN: Recurrent Neural Network
- RV: Realised Variance
- SGD: Stochastic Gradient Descent
- SHAP: SHapley Additive exPlanations
- SPA: Superior Predictive Ability (test)
- TCN: Temporal Convolutional Network
- TFT: Temporal Fusion Transformer
- TSA: Time Series Analysis
- US: United States
- VC: Vapnik–Chervonenkis (dimension)
- VIX: CBOE Volatility IndeX
- WIG20: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)



Today's question and route

- **Question:** when do flexible learners (penalised regressions, tree ensembles, deep networks) forecast time series better than strong statistical baselines, and how do we know it honestly?
 - ▶ dependence breaks the i.i.d. logic of machine learning: validation, generalisation and inference all change
- **Route** of the chapter
 - ▶ learning from dependent data: mixing, bounds, when cross-validation is valid
 - ▶ global and local models; lasso and its relatives in high dimension; tree ensembles in depth
 - ▶ recurrent and convolutional networks; attention and Transformers; N-BEATS, N-HiTS, DeepAR, TFT
 - ▶ interpretation and the honest benchmark: baselines, tests, data snooping
- We build on [TSA, Chapter 9](#) (lag features, walk-forward validation, trees, a small MLP, M4 and M5) and on [Chapter 1](#) (DM, MCS, scoring rules); Seminar 12 comes before this lecture

Learning outcomes

- ▣ State when K-fold cross-validation is valid for autoregressive prediction, and design blocked, purged and rolling-origin validation that avoids leakage
- ▣ Derive the lasso, adaptive lasso and elastic net solutions, state their consistency conditions for dependent data, and explain why naive inference after selection fails
- ▣ Explain the bias–variance mechanics of bagging, random forests, quantile regression forests and gradient boosting with constraints
- ▣ Derive vanishing gradients in recurrent networks and the role of gates; read the architectures of TCN, Transformers, N-BEATS, N-HiTS, DeepAR and TFT
- ▣ Run a pre-registered comparison against strong baselines with DM, MCS and seeds reported, and interpret Shapley values and attention correctly

Reading, data and tools

- ▣ Validation and learning theory: [Bergmeir, Hyndman and Koo \(2018\)](#); [Racine \(2000\)](#); [Mohri and Rostamizadeh \(2010\)](#); [Hewamalage, Ackermann and Bergmeir \(2023\)](#)
 - ▶ High dimension and trees: [Tibshirani \(1996\)](#); [Zou \(2006\)](#); [Basu and Michailidis \(2015\)](#); [Meinshausen \(2006\)](#); [Friedman \(2001\)](#); [Ke et al. \(2017\)](#)
 - ▶ Deep learning: [Hochreiter and Schmidhuber \(1997\)](#); [Vaswani et al. \(2017\)](#); [Zeng et al. \(2023\)](#); [Oreshkin et al. \(2020\)](#); [Salinas et al. \(2020\)](#); [Lim et al. \(2021\)](#); [survey Benidis et al. \(2022\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_12](#)
 - ▶ `numpy`, `scikit-learn` and `PyTorch` on a CPU: small networks, fixed seeds, runs of minutes
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source, sample	Use
Romanian electricity load, hourly	Energy-Charts (ENTSO-E transparency data), 1 369 days to 30 September 2026	trees, Transformers, DLinear
HICP inflation, 27 EU countries	Eurostat prc_hicp_minr, annual rates, monthly, January 2000 – August 2026	global models, lasso
S&P 500 and five other indices, realised variance	Heber et al. (2009) : 5-minute RV, 5 552 days to 25 February 2022	deep models against HAR
M4 hourly subset	Makridakis, Spiliotis and Assimakopoulos (2020) : 414 series of 700–960 hours, horizon 48	N-BEATS, N-HiTS, DeepAR

- Every comparison has a forecast period fixed before the models are run; losses, tests and seeds are reported with it

Four generations of forecasting models



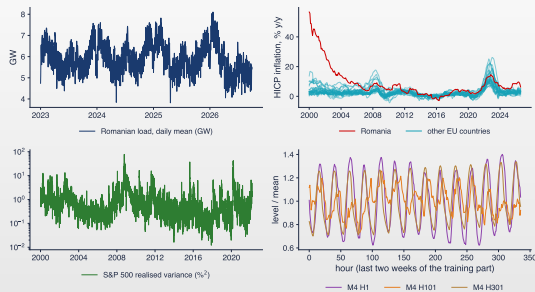
Geoffrey Hinton, Nobel Prize in Physics 2024

Photo: Arthur Petron (2024); CC BY-SA 4.0;

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- **Statistical:** ARIMA, ETS, HAR, state space: few parameters, one model per series
- **Machine learning:** penalised regressions, forests, boosting on lag features, often global
- **Deep sequence models:** RNN, LSTM, TCN, Transformers, N-BEATS, DeepAR; backpropagation [Rumelhart, Hinton and Williams \(1986\)](#)
- **Foundation models:** pretrained on many series, used zero-shot ([Chapter 13](#))
- Each generation must beat the previous one out of sample, with tests, on the same data

Four data sets, four kinds of structure



- Romanian load (daily mean, 6.06 GW on average); EU inflation (Romania in red); S&P 500 realised variance (log scale); three M4 hourly series (last two weeks)

Interpreting the four data sets

- ▣ Load: strong daily and weekly seasonality, holidays, slow annual cycle: calendar structure that trees and linear models capture well
- ▣ Inflation: 27 short, co-moving series with a common shock in 2021–2023: the natural ground for global models and for the high-dimensional panel
- ▣ Realised variance: long memory and heavy tails; the HAR benchmark of [Chapter 8](#) is hard to beat
- ▣ M4 hourly: many related series of the same frequency, the setting where global deep models were first shown to win

Known from TSA, and new here

- ▣ Known (TSA, Chapter 9): lag features, walk-forward validation, leakage by example, ridge and lasso, trees, random forests, gradient boosting, a small MLP, conformal basics, M4 and M5
- ▣ New: the statistics behind the recipes
 - ▶ generalisation under mixing; when cross-validation is valid; global models
 - ▶ selection consistency and post-selection inference; quantile forests; constrained boosting
 - ▶ gradients through time; attention; basis expansion; likelihood-based deep models; interpretation
- ▣ Replications: Bergmeir, Hyndman and Koo (2018, Section 4); Montero-Manso and Hyndman (2021) on EU inflation; Zeng et al. (2023) protocol on Romanian load; the M4 hourly evaluation

Risk, empirical risk and the generalisation gap

- ▣ Forecast rule $f \in \mathcal{F}$, loss ℓ ; **risk** $R(f) = \mathbb{E} \ell(Y_{t+h}, f(X_t))$ under the stationary law; **empirical risk** $\hat{R}_n(f) = \frac{1}{n} \sum_t \ell(y_{t+h}, f(x_t))$
 - ▶ ERM: $\hat{f} = \arg \min_{\mathcal{F}} \hat{R}_n$; the gap $R(\hat{f}) - \hat{R}_n(\hat{f}) \leq \sup_{\mathcal{F}} |R - \hat{R}_n|$
- ▣ i.i.d. data: $\sup_{\mathcal{F}} |R - \hat{R}_n| = O_p(\sqrt{\text{complexity}(\mathcal{F})/n})$ (VC dimension, Rademacher complexity)
- ▣ Dependent data: the n observations carry less information; the bound holds with an **effective sample size** smaller than n
 - ▶ and only if the series forgets its past fast enough: mixing
- ▣ Non-stationarity (breaks, [Chapter 2](#)) adds a second gap: the future law differs from the training law; no bound covers it

Mixing coefficients

- β -mixing: $\beta(a) = \sup_t \mathbb{E} \sup_{B \in \sigma(X_{t+a}, \dots)} |P(B \mid \sigma(\dots, X_t)) - P(B)|$; α -mixing uses $|P(A \cap B) - P(A)P(B)|$
 - ▶ mixing: $\beta(a) \rightarrow 0$ as the gap $a \rightarrow \infty$: the far future is almost independent of the past
- Stationary ARMA with continuous innovations, GARCH and Markov-switching models with ergodic chains are **geometrically** β -mixing: $\beta(a) \leq Cr^a$
 - ▶ long memory ([Chapter 10](#)) is not geometrically mixing: the effective sample size shrinks much more
- Mixing is assumed, rarely tested: it is a statement about the data-generating process, not about the sample

Blocking and a generalisation bound

- **Yu (1994)**: split the sample into 2μ alternating blocks of length a ; under β -mixing the odd blocks behave like μ **independent** blocks up to an error $\mu\beta(a)$
 - ▶ i.i.d. tools then apply with $\mu = n/(2a)$ observations instead of n
- **Mohri and Rostamizadeh (2010)**: for a uniformly stable learner (stability κ_n), with probability $\geq 1 - \delta$:
$$R(\hat{f}) \leq \hat{R}_n(\hat{f}) + O(\kappa_n a) + O(\sqrt{\log(1/\delta')/\mu}), \delta' = \delta - 2(\mu - 1)\beta(a)$$
 - ▶ trade-off in a : long blocks make $\beta(a)$ small but leave few blocks μ
 - ▶ ridge, kernel ridge and regularised models are uniformly stable; deep networks trained by SGD only approximately
- The same blocking idea gives blocked bootstrap ([Chapter 0](#)) and blocked cross-validation

Cross-validation for time series: the schemes

- ▣ **Random K-fold**: rows of the lag matrix $(y_t; y_{t-1}, \dots, y_{t-p})$ assigned to folds at random
- ▣ **Non-dependent CV** Bergmeir, Hyndman and Koo (2018): the same folds, but training rows within p lags of a test row are removed
- ▣ **Blocked K-fold** and **hv-block** Burman, Chow and Nolan (1994), Racine (2000): contiguous test blocks, a gap of h rows (purge) on each side
- ▣ **Out-of-sample (OOS)** and **rolling origin**: test only on the end; the only scheme that mimics real use, at the price of a high variance
- ▣ The question is not which scheme is “correct” but which estimates the future risk with small bias and small variance

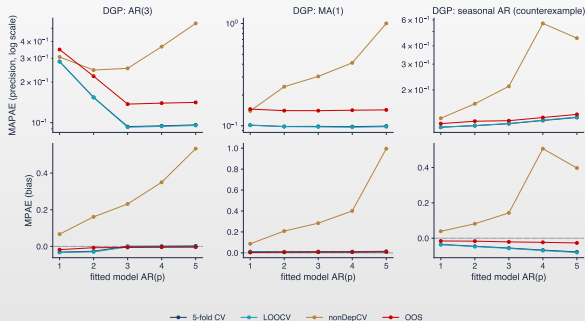
When K-fold is valid: Bergmeir, Hyndman and Koo (2018)

- Model $y_t = g(y_{t-1}, \dots, y_{t-p}; \theta) + \varepsilon_t$ with **uncorrelated** errors ε_t [Bergmeir, Hyndman and Koo \(2018\)](#)
 - ▶ the rows of the lag matrix are then exchangeable enough: a test row shares no error with a training row
 - ▶ result: the K-fold estimate of the MSE is asymptotically unbiased, and has a lower variance than OOS because every row is used
- It fails when the residuals are autocorrelated: an under-specified model, overlapping h -step targets, or features that identify time
 - ▶ then a test error is predictable from the neighbouring training errors: optimistic estimates
- Practical rule: check the residual ACF of the fitted model (Ljung–Box, [TSA, Chapter 2](#)) before trusting random K-fold

Replication design: Bergmeir, Hyndman and Koo (2018, Section 4)

- ▣ Series of length 200: in-set 140, out-set 60; models AR(1)–AR(5) by OLS; one-step forecasts
- ▣ Procedures on the in-set: 5-fold CV, LOOCV, non-dependent CV ($p = 5$), OOS (last 20%); “true” error: the model on the whole in-set, RMSE on the out-set
- ▣ 1 000 trials per DGP: stationary AR(3) with random coefficients, invertible MA(1), and a seasonal AR fitted to US accidental deaths 1973–1978 ($\phi = 0.76$, $\Phi_{12} = 0.85$) as the counterexample
- ▣ Precision MAPAE = $\text{mean}|\widehat{PE} - PE|$, bias MPAAE = $\text{mean}(\widehat{PE} - PE)$

Which validation estimates the future error?



Top: precision (log scale); bottom: bias, by fitted AR order and data-generating process

Interpreting the replication

- ▣ AR(3) data, AR(3) model: 5-fold CV has MAPAE 0.094 against 0.137 for OOS, bias 0.001: as in the paper, CV is unbiased and more precise
- ▣ Under-specified AR(1) on AR(3) data: residuals are autocorrelated and CV loses its edge (MAPAE 0.284 against 0.349)
- ▣ Seasonal counterexample: every $\text{AR}(p \leq 5)$ misses lag 12; CV is biased down (MAPAE -0.077 for AR(5)) and no more precise than OOS (0.130 against 0.136)
- ▣ Non-dependent CV removes so many rows that its models are poor: bias 0.232 for AR(3); the price of independence is paid in data

Leakage through overlapping targets and time features

- Target $z_t = \frac{1}{h} \sum_{j=1}^h y_{t+j}$ (a month of volatility, a quarter of sales): $\text{corr}(z_t, z_{t+1}) \geq (h-1)/h$ when y is white noise, more when y is persistent
 - ▶ a random fold puts z_{t-1} and z_{t+1} in training when z_t is tested: the answer is almost in the training set
- A time index or a date feature lets a flexible learner **locate** the test row between its neighbours
- Simulation: AR(1) with $\phi = 0.9$, $h = 20$, a random forest on $y_t, \dots, y_{t-4}$ and a time index; 60 replications; ratio of the estimated to the true RMSE on new data

How optimistic is each validation scheme?



- Ratio 1: the scheme estimates the error on new data without bias; below 1: optimistic

Interpreting the leakage experiment

- ▣ Random 5-fold: median ratio 0.34 (quartiles 0.28–0.39): it claims an error three times smaller than the truth
- ▣ Blocked 5-fold 0.92, purged 5-fold 0.98, last 20% 0.98: contiguity and the purge remove most of the bias
- ▣ The interior blocks still let the forest interpolate in time; only the last block reproduces extrapolation into the future
- ▣ Rule: purge at least h rows around every test block and drop time indices unless the trend is modelled explicitly

Recap: Learning from dependent data

- Generalisation bounds hold with an effective sample size set by mixing; blocking is the common tool
- Random K-fold is valid for autoregressions with uncorrelated errors, and then better than OOS
- Overlapping targets, under-fitting and time features break it: block, purge, and keep a final untouched period

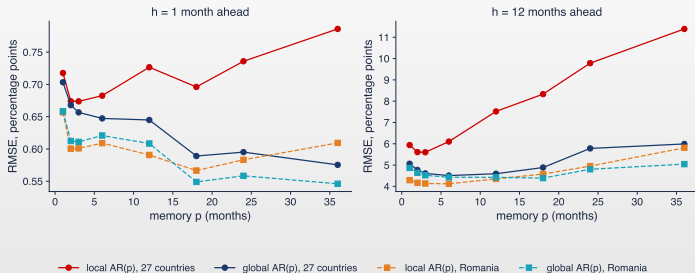
Local and global models

- **Local**: one model per series, $\hat{y}_{i,t+h} = f_i(y_{i,t}, y_{i,t-1}, \dots)$; **global**: one function for all series, $\hat{y}_{i,t+h} = f(y_{i,t}, y_{i,t-1}, \dots)$
 - ▶ global models pool $N \times T$ rows: far more data per parameter
- **Montero-Manso and Hyndman (2021)**: for any set of series a global model exists that forecasts as well as the local ones; global models can afford **longer memory** and more complexity
 - ▶ their bound: the complexity of a global model grows with NT , that of N local models with T each
- The M4 and M5 winners were global (**Smyl (2020)**; LightGBM in **Makridakis, Spiliotis and Assimakopoulos (2022)**): pooling, not depth, was the common ingredient
- Scaling matters: series of different levels are divided by their own scale before pooling (inflation rates share a scale)

Design: EU inflation, local against global

- ▣ Annual HICP inflation of the 27 EU countries; direct forecasts $h = 1$ and $h = 12$ months ahead from $AR(p)$ models, $p \in \{1, 2, 3, 6, 12, 18, 24, 36\}$
- ▣ Rolling window of 120 months; forecast origins from January 2015 (139 origins for $h = 1$)
- ▣ Local: 27 OLS regressions per origin; global: one pooled OLS (the same coefficients for all countries), as in [Montero-Manso and Hyndman \(2021\)](#)
- ▣ Loss: RMSE in percentage points; DM test on the cross-country average loss differential (HLN correction, [Chapter 1](#))

Memory helps global models and hurts local ones



□ RMSE against the memory p ; circles: average over the 27 countries; squares: Romania

Interpreting local and global

- ▣ $h = 1$: local RMSE is best at $p = 3$ (0.67) and then deteriorates; global keeps improving up to $p = 36$ (0.58); DM on the 27-country average -4.40 ($p < 0.001$)
- ▣ $h = 12$: the 2021–2023 surge dominates; local AR(36) reaches 11.39 against 5.99 global; best global 4.51 against best local 5.60, DM -1.09 (p 0.277)
- ▣ Romania alone: the global model helps at $h = 1$ with long memory (0.55 against 0.61 at $p = 36$), not significantly (DM p 0.19)
- ▣ Lesson of Montero-Manso and Hyndman confirmed: pooling buys memory; for a single series the gain is real but hard to prove

The lasso: optimality conditions

- **Tibshirani (1996)**: $\hat{\beta} = \arg \min_{\beta} \frac{1}{2n} \|y - X\beta\|^2 + \lambda \|\beta\|_1$; KKT: $\frac{1}{n} x_j'(y - X\hat{\beta}) = \lambda \text{sign}(\hat{\beta}_j)$ if $\hat{\beta}_j \neq 0$, $\leq \lambda$ in absolute value otherwise
 - ▶ orthonormal design ($X'X = nI$): $\hat{\beta}_j = \text{sign}(z_j)(|z_j| - \lambda)_+$, soft thresholding of the OLS coefficient z_j
 - ▶ ridge: $\hat{\beta}_j = z_j/(1 + \lambda)$: shrinks, never selects
- Prediction: with s non-zero coefficients and a restricted eigenvalue κ , $\frac{1}{n} \|X(\hat{\beta} - \beta)\|^2 = O_p(s \log p / (n\kappa^2))$ for $\lambda \asymp \sigma \sqrt{\log p / n}$
 - ▶ time series: the same rate for stable VARs, with κ depending on the spectral density bounds [Basu and Michailidis \(2015\)](#); heavy-tailed, heteroskedastic errors [Medeiros and Mendes \(2016\)](#)
- Selection is harder than prediction: sign consistency needs the **irrepresentable condition** $\|X'_{S^c} X_S (X'_S X_S)^{-1} \text{sign}(\beta_S)\|_{\infty} < 1$ [Zhao and Yu \(2006\)](#): it fails when relevant and irrelevant predictors are correlated (neighbouring countries, lags)

Adaptive lasso and elastic net

- ▣ **Adaptive lasso** [Zou \(2006\)](#): penalty $\lambda \sum_j w_j |\beta_j|$, $w_j = 1/|\tilde{\beta}_j|^\gamma$ from a first-step estimate (OLS, ridge)
 - ▶ large coefficients are penalised little, small ones a lot: the **oracle property** (consistent selection and $\sqrt{n}$ -normal estimates of the non-zero coefficients) without the irrerepresentable condition
- ▣ **Elastic net** [Zou and Hastie \(2005\)](#): $\lambda(\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2)$; orthonormal design:
 $\hat{\beta}_j = \text{sign}(z_j)(|z_j| - \lambda\alpha)_+ / (1 + \lambda(1 - \alpha))$
 - ▶ grouping effect: correlated predictors enter together instead of one being chosen at random
- ▣ The penalty is tuned by CV: with h -step targets use blocked folds with a purge of h periods (Section 1)

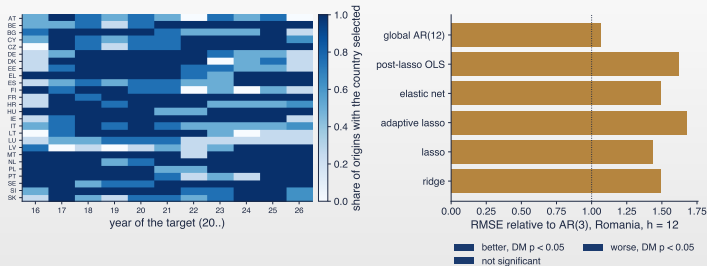
Inference after selection

- Naive practice: select with the lasso, then report OLS t -statistics of the selected variables: the p -values are **too small**
 - ▶ the same data chose the model and test it: the distribution of $\hat{\beta}$ is conditional on being selected
- Remedies: **PoSI** intervals valid for any selection [Berk et al. \(2013\)](#); **exact conditional** intervals for the lasso (a polyhedral selection event) [Lee et al. \(2016\)](#)
- For one coefficient of interest (a policy rate, a foreign shock): **double selection** [Belloni, Chernozhukov and Hansen \(2014\)](#): lasso of y on the controls, lasso of the regressor of interest on the controls, OLS on the union
 - ▶ with HAC standard errors for time series ([Chapter 0](#))
- For forecasting, selection is a means: judge it by out-of-sample loss and by the stability of what is selected

Design: Romanian inflation from the EU panel

- Target: Romanian annual HICP inflation 12 months ahead; window 120 months, re-estimation every 3 months; 128 forecasts (January 2016 – August 2026)
- Features: 3 own lags (never penalised) and 3 lags of the 26 other countries: 78 penalised predictors for about 105 observations
- Models: AR(3) (benchmark), ridge, lasso, adaptive lasso, elastic net ($\alpha = 0.5$), post-lasso OLS, global AR(12) of the panel; penalty by blocked 5-fold CV with a 12-month purge
- In the spirit of the data-rich inflation literature [Medeiros et al. \(2021\)](#), [Goulet Coulombe et al. \(2022\)](#), with the EU panel as the data set

What the panel lasso selects, and what it delivers



- Left: share of forecast origins in each year at which a country enters the lasso; right: RMSE relative to AR(3), colour by the DM test

Interpreting the panel lasso

- ▣ Selection is unstable: median 34 of 78 predictors (from 15 to 61); the most frequent are EL, NL, MT, not Romania's neighbours
- ▣ Accuracy: AR(3) RMSE 4.13; lasso 1.44, adaptive lasso 1.67, elastic net 1.49, ridge 1.49 times as large; global AR(12) 1.06
- ▣ None of the differences is significant (DM p from 0.20 to 0.39): about 10 independent 12-month periods, one dominated by the 2022 surge
- ▣ High dimension with 105 observations of a persistent target: CV picks small penalties, the panel fits past co-movement that does not repeat; parsimony wins here

Recap: Global models and high dimension

- Global models pool many series and can afford long memory; on EU inflation they win at one month
- The lasso predicts well under sparsity and restricted eigenvalues; it selects well only under the irrepresentable condition; adaptive lasso and elastic net relax it
- Report selection stability and use PoSI, exact or double-selection inference, never naive t -statistics

Bagging: the variance of an average

- B predictors $\hat{f}_b(x)$, identically distributed, variance σ^2 , pairwise correlation ρ (over the data and the randomisation) [Breiman \(1996\)](#)
 - ▶ $\text{Var}(\frac{1}{B} \sum_b \hat{f}_b) = \frac{1}{B^2} (B\sigma^2 + B(B-1)\rho\sigma^2) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$
 - ▶ more trees remove only the second term; the floor $\rho\sigma^2$ is lowered only by **decorrelating** the trees
- **Random forests** [Breiman \(2001\)](#): a random subset of m_{try} features at each split lowers ρ at the cost of a little bias
- Theory: consistency of Breiman's forests for additive regression [Scornet, Biau and Vert \(2015\)](#); honest trees (separate samples for splits and leaf values) give asymptotically Normal forecasts with variance estimates [Wager and Athey \(2018\)](#)
- Trees cannot extrapolate: a forecast is an average of past targets, so trends and level shifts need differencing or a linear part

Quantile regression forests

- ▣ [Meinshausen \(2006\)](#): keep **all** training targets in the leaves; weights $w_i(x) = \frac{1}{B} \sum_b \frac{1_{\{x_i \in L_b(x)\}}}{|L_b(x)|}$, $\sum_i w_i(x) = 1$
 - ▶ conditional CDF $\hat{F}(y | x) = \sum_i w_i(x) 1\{y_i \leq y\}$; quantile $\hat{q}_\tau(x) = \inf\{y : \hat{F}(y | x) \geq \tau\}$
- ▣ The same forest gives every quantile: no crossing, no refit per level
 - ▶ consistent for the conditional distribution under regularity conditions; a random forest is an adaptive nearest-neighbour method
- ▣ Evaluation with the pinball loss and coverage ([Chapter 1](#)): the bands are honest only if the forest was trained on data like the future

Gradient boosting as functional gradient descent

- ▣ **Friedman (2001)**: $F_m = F_{m-1} + \nu h_m$, h_m a small tree fitted to the negative gradient $g_i = -\partial \ell(y_i, F) / \partial F|_{F_{m-1}(x_i)}$
 - ▶ squared loss: $g_i = \text{residual}$; pinball loss: $g_i = \tau - 1\{y_i < F\}$ (quantile boosting); shrinkage ν trades steps for variance
- ▣ Second order **Chen and Guestrin (2016)**: with gradients G and Hessians H summed in a leaf, the optimal leaf value is $w^* = -G / (H + \lambda)$ and a split gains $\frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda} \right] - \gamma$
- ▣ **LightGBM Ke et al. (2017)**: features binned into histograms (at most 255 bins), leaf-wise growth, gradient-based one-side sampling; the scikit-learn HistGradientBoosting used here follows the same design
- ▣ **Monotone constraints**: a split is allowed only if the left and right leaf values respect the order, and children inherit bounds; the fit is then monotone in that feature everywhere

Design: Romanian day-ahead load



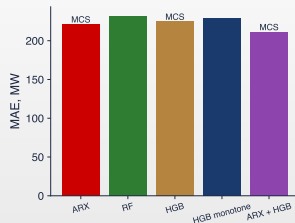
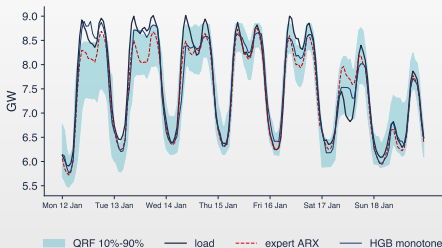
750 kV line at Mărașești, Romania

Photo: TrainSimFan (2022); CC BY-SA 4.0;

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- Forecast the 24 hourly loads of day $d + 1$ with data up to day d ; evaluation 638 days, January 2025 – September 2026
- Benchmark: the expert ARX of [Chapter 1 \(Ziel and Weron \(2018\)\)](#): one regression per hour, 364-day window, refitted daily
- Global learners (one model for all hours) on the same lags, hour, weekday, Romanian holidays, annual cycle; refitted monthly on all earlier data: RF with QRF bands, HGB, HGB increasing in the three load lags, quantile HGB
- Losses: MAE, average pinball over the deciles, 80% coverage; DM against ARX and MCS ([Chapter 1](#))

Trees against the expert ARX

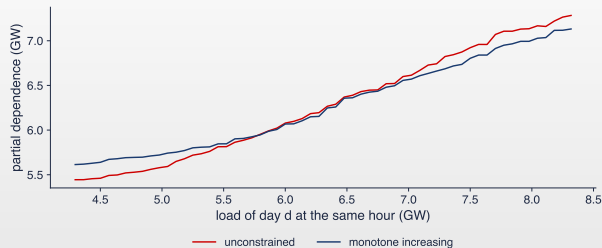


- Left: one January week with QRF 10%–90% bands; right: MAE in MW, MCS marks the models in the 90% Model Confidence Set

Interpreting the load comparison

- ▣ MAE (MW): ARX 221, RF 232, HGB 225, monotone HGB 228; the forest is significantly worse (DM 2.84), boosting is not significantly different from ARX
- ▣ The equal-weight average of ARX and boosting: 211 MW, DM -3.36 ($p < 0.001$), the only model in the MCS: combination beats selection ([Chapter 1](#))
- ▣ Quantiles: pinball (MW) ARX 89.1, quantile HGB 89.5, QRF 92.8; 80% coverage 78%, 61%, 83%: quantile boosting is sharp but under-covers, QRF over-covers
- ▣ A daily-refitted linear model with the right structure is a strong baseline; trees add value as a complement, not as a replacement

Monotone constraints in practice



- Partial dependence of the boosted forecast on the load of day d at the same hour, trained on 2023–2024

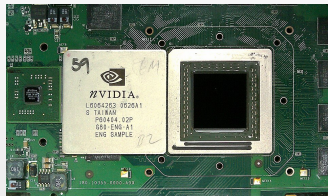
Interpreting the constraint

- Unconstrained: 3 decreasing steps (the largest 7.5 MW): more load yesterday would sometimes mean less load tomorrow, a fit to noise
- Constrained: monotone by construction; total range 1.52 GW against 1.84 GW: part of the effect moves to the other lags
- Accuracy barely changes (228 against 225 MW); the gain is plausibility and robustness at the edges of the data
- Constraints are how economic knowledge enters a black box: sign restrictions, as in [Chapter 3](#), but for nonlinear learners

Recap: Tree ensembles

- ▣ Bagging and forests reduce variance down to the floor $\rho\sigma^2$; decorrelation, not the number of trees, lowers it
- ▣ QRF gives the whole conditional distribution from one forest; boosting fits gradients, with second-order leaves and histograms in LightGBM
- ▣ On Romanian load, trees match but do not beat the expert ARX; their combination wins

From the MLP to sequence models



NVIDIA G80, the first CUDA graphics processor
(2006)

Photo: Hyins (2011); public domain; Wikimedia
Commons

- An MLP on a window of L lags is a nonlinear $AR(L)$:
 $\hat{y}_{t+h} = W_2 \sigma(W_1 x_t + b_1) + b_2$, $\sigma = \text{ReLU}$ or $\tanh$
- Training: backpropagation [Rumelhart, Hinton and Williams \(1986\)](#) with Adam [Kingma and Ba \(2015\)](#); early stopping on the last block of the training sample; inputs standardised with training moments only
- Universal approximation says nothing about data needs: with a few thousand daily observations a network has little room over a linear model
- Sequence models share weights across time: recurrence (RNN), convolution (TCN) or attention (Transformer)

Recurrent networks and backpropagation through time

- Elman RNN: $h_t = \tanh(W h_{t-1} + U x_t + b)$, $\hat{y}_t = V h_t$; the state h_t summarises the past, like the Kalman state (Chapter 6) but learned
- BPTT: $\frac{\partial \mathcal{L}_T}{\partial W} = \sum_{t \leq T} \frac{\partial \mathcal{L}_T}{\partial h_T} \left(\prod_{k=t+1}^T \frac{\partial h_k}{\partial h_{k-1}} \right) \frac{\partial^+ h_t}{\partial W}$, $\frac{\partial h_k}{\partial h_{k-1}} = \text{diag}(1 - h_k^2) W$
 - ▶ the contribution of lag $T - t$ is a product of $T - t$ Jacobians
- Bengio, Simard and Frasconi (1994), Pascanu, Mikolov and Bengio (2013): $\| \prod_k \partial h_k / \partial h_{k-1} \| \leq (\gamma \|W\|)^{T-t}$ with $\gamma = \max |\tanh'| \leq 1$
 - ▶ $\gamma \|W\| < 1$: **vanishing** gradients, long lags cannot be learned; spectral radius > 1 : possible **explosion**, cured by gradient clipping

LSTM and GRU: gates



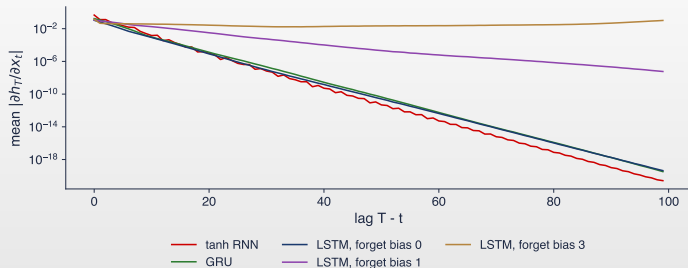
Sepp Hochreiter, co-author of the LSTM

Photo: Eulenreich (2015); CC BY-SA 4.0;

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- LSTM Hochreiter and Schmidhuber (1997): gates
 $i_t, f_t, o_t = \sigma(W_\bullet[h_{t-1}, x_t] + b_\bullet)$, candidate $\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$
- cell $c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$, output $h_t = o_t \odot \tanh(c_t)$; along the cell, $\partial c_t / \partial c_{t-1} = \text{diag}(f_t)$: additive, no squashing
- The forget gate Gers, Schmidhuber and Cummins (2000) must stay open: $f \approx 1$; a forget bias of 1 is the usual initialisation Jozefowicz, Zaremba and Sutskever (2015)
- GRU Cho et al. (2014): update z_t and reset r_t gates,
 $h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$: fewer parameters, similar behaviour

How far back does the gradient reach?



□ Mean $|\partial h_T / \partial x_t|$ of freshly initialised networks (32 units, $T = 100$, Gaussian inputs, average of 5 seeds), log scale

ATS_ch12_recurrent

Interpreting the gradients

- ▣ At lag 50 the gradient is a fraction 10^{-10} of its lag-1 value for the tanh RNN, 10^{-9} for the GRU and 10^{-9} for the LSTM with the default forget bias 0
- ▣ Forget bias 1: 10^{-4} ; forget bias 3: 0.47: gates help only when the forget gate is open ($\sigma(3) = 0.95$)
- ▣ Decay rates per lag: 0.51 (RNN), 0.45 (LSTM, bias 0), 0.03 (LSTM, bias 3): geometric in every case
- ▣ For daily series this matters less than it seems: HAR-type memory (22 lags) is within reach; for hourly data with weekly cycles (168 lags) it decides the architecture

Temporal convolutional networks

- Causal convolution: $z_t = \sum_{j=0}^{k-1} w_j x_{t-dj}$ uses only the past; dilation d skips $d - 1$ steps [van den Oord et al. \(2016\)](#)
 - ▶ TCN [Bai, Kolter and Koltun \(2018\)](#): residual blocks of two dilated causal convolutions, $d = 1, 2, 4, \dots, 2^{\ell-1}$
- Receptive field: $R = 1 + 2(k - 1)(2^\ell - 1)$; $k = 3$, $\ell = 4$: $R = 61$; one week of hours needs $\ell = 7$ ($R = 509$)
 - ▶ memory is set by the architecture, not learned: no vanishing over time, but nothing beyond R
- Parallel over time (fast training), stable gradients; Bai et al. found TCNs competitive with LSTMs on standard sequence tasks

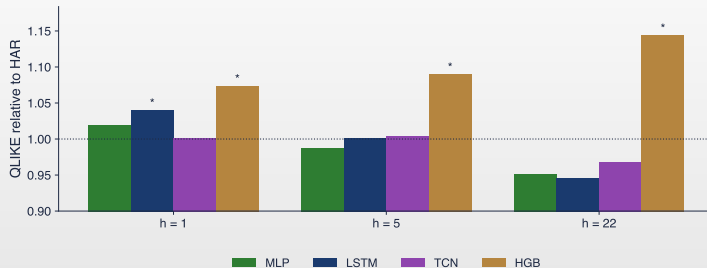
Multi-step forecasts and sequence-to-sequence

- ▣ **Recursive:** one-step model iterated, forecasts fed back as inputs: errors accumulate, the model is trained for $h = 1$ only
- ▣ **Direct:** one model per horizon h (our HAR and network forecasts); **MIMO:** one network outputs the whole vector $(\hat{y}_{t+1}, \dots, \hat{y}_{t+H})$ (DLinear, N-BEATS, Transformers here)
- ▣ **Encoder-decoder** [Sutskever, Vinyals and Le \(2014\)](#): an encoder RNN reads the past into a state, a decoder RNN unrolls the future; trained with teacher forcing
 - ▶ exposure bias: at test time the decoder sees its own errors; DeepAR samples paths instead
- ▣ Survey of recurrent forecasting practice [Hewamalage, Bergmeir and Bandara \(2021\)](#): deseasonalise or add seasonal lags, scale per series, train globally, ensemble seeds

Design: deep models against HAR

- ▣ Target $y = \log RV$ of the S&P 500 (Heber et al. (2009)); forecasts of RV_{t+h} , $h \in \{1, 5, 22\}$, direct; level forecast $\exp(\hat{m} + s^2/2)$ with the training residual variance
- ▣ Expanding window, first forecast after 2500 days (24 December 2009), re-estimation every 500 days; 3 051 forecasts per horizon
- ▣ Models on the last 22 values: HAR Corsi (2009) (OLS), MLP (2×32 units), LSTM (16 units), TCN (3 levels, $R = 29$), HGB on the 22 lags; one seed per network
- ▣ QLIKE Patton (2011), DM (HLN) against HAR, MCS; the literature: neural networks gain little over HAR Bucci (2020); ML gains appear mostly with many predictors Christensen, Siggaard and Veliyev (2023)

Can deep networks beat HAR?



□ QLIKE relative to HAR (1 = HAR); a star marks a DM rejection at 5%

Q ATS_ch12_recurrent

Interpreting deep models against HAR

- ▣ $h = 1$: MLP 1.019, LSTM 1.040 (DM 2.77), TCN 1.002, HGB 1.074 (DM 3.84): nothing beats HAR, two are significantly worse
- ▣ $h = 22$: MLP 0.951, LSTM 0.946, TCN 0.968: gains of 3–5% that DM does not confirm (p 0.132, 0.444); HAR stays in the MCS (p 0.49)
- ▣ Boosting on raw lags is the worst at every horizon: trees cannot extrapolate to the volatility peaks of 2020
- ▣ HAR is a linear model with the right three features; a network must rediscover them from a few thousand days

Recap: Deep learning for sequences

- ▣ Gradients through time are products of Jacobians: they vanish geometrically unless the LSTM forget gate stays open
- ▣ TCNs fix the memory by the receptive field; multi-step forecasts are recursive, direct or MIMO
- ▣ On daily realised variance, networks do not significantly beat HAR at any horizon

Scaled dot-product attention

- Vaswani et al. (2017): tokens $X \in \mathbb{R}^{n \times d}$; $Q = XW_Q$, $K = XW_K$, $V = XW_V$;
 $\text{Att}(Q, K, V) = \text{softmax}(QK'/\sqrt{d_k})V$
 - ▶ each output is a weighted average of the values, with data-dependent weights; $\sqrt{d_k}$ keeps the logits of order one
 - ▶ multi-head: several projections in parallel, concatenated
- Permutation equivariant: without **positional encodings** the order of the tokens is lost, which is fatal for time series
- Cost $O(n^2d)$ in time and memory: one token per hour and a week of input give $n = 168$; a month gives $n = 720$
 - ▶ the long-horizon Transformers of 2020–2022 were mostly about this cost

Transformers for long horizons

- **Informer** Zhou et al. (2021): ProbSparse attention (only the most informative queries), self-attention distilling, a generative decoder that outputs the horizon at once: $O(n \log n)$
- **Autoformer** Wu et al. (2021): series decomposition inside every layer (moving-average trend and remainder) and an auto-correlation mechanism that aggregates sub-series at lags found by FFT
- **PatchTST** Nie et al. (2023): tokens are **patches** of consecutive values (e.g. 16 or 24), each channel modelled separately, instance normalisation
 - ▶ fewer tokens (n/P), local semantics in each token, longer look-backs affordable
- All were evaluated on the same handful of benchmarks (electricity, traffic, weather, ETT) with MSE on standardised data

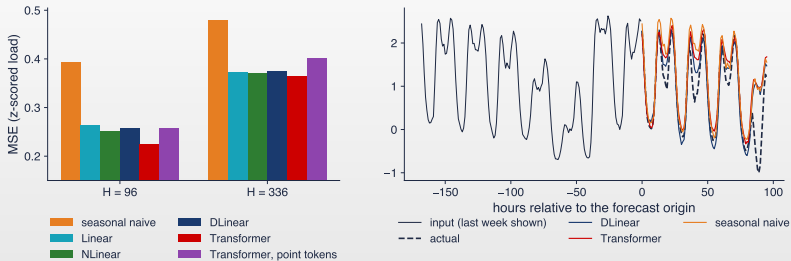
Are Transformers effective? The Zeng et al. (2023) critique

- Zeng et al. (2023): self-attention is permutation-invariant; point-wise tokens carry no local semantics; positional encodings only partly restore order
- LTSF-Linear baselines: **Linear** $\hat{y}_{t+1:t+H} = Wx_{t-L+1:t}$; **NLinear** subtracts and adds back the last value; **DLinear** splits x into a moving-average trend and a remainder, one linear layer each
 - ▶ DLinear is itself one linear map: $\hat{y} = (W_s(I - A) + W_t A)x$, A the moving-average matrix (Seminar 12, A8)
- Their finding: these one-layer models beat Informer, Autoformer and their successors on most of the standard long-horizon benchmarks
 - ▶ and the Transformers did not improve with a longer look-back L : a sign they did not use the extra history
- PatchTST was the reply: patching and channel independence put Transformers back ahead of DLinear on the same benchmarks

Replication design: the Zeng et al. protocol on Romanian load

- Hourly Romanian load as one series ($n = 32\,856$ hours), z-scored with training moments; chronological split 7:1:2; test from 31 December 2025
- Look-back $L = 336$ hours; horizons $H = 96$ and 336 ; MSE and MAE on the z-scored series, as in their tables
- Models: repeat the last value, seasonal naive (last week), Linear, NLinear, DLinear (kernel 25), a patch Transformer (14 patches of 24 hours, 1 layer, 4 heads) and the same Transformer with one token per hour
- Adam, MSE, early stopping on the validation block; three seeds averaged (one for the point-token Transformer, whose cost is $O(n^2)$)

Linear models, Transformers and the role of tokens



- Left: test MSE by horizon (repeating the last value, 1.585 and 1.699, is off the scale); right: one test window at $H = 96$ (the last week of input, the actual load and three forecasts)

Interpreting the protocol

- $H = 96$: seasonal naive 0.393, Linear 0.265, NLinear 0.253, DLinear 0.257; point-token Transformer 0.258; patch Transformer 0.226
- $H = 336$: DLinear 0.374, point tokens 0.402 (worse than every linear model), patches 0.364
- Both papers replicate on our data: the point-token Transformer is no better than NLinear or DLinear at $H = 96$ and worse than every linear model at $H = 336$; patching overturns the ranking
- The gap between the best linear model and the patch Transformer shrinks from 0.253 against 0.226 at $H = 96$ to 0.371 against 0.364 at $H = 336$: at long horizons the series is mostly seasonal level

The Temporal Fusion Transformer

- Lim et al. (2021): built for multi-horizon forecasts with three kinds of inputs: static (the store, the country), known future (calendar, prices), observed past
- Components
 - ▶ gated residual networks (GRN): $\text{LayerNorm}(a + \text{GLU}(\eta))$, a gate that can switch a block off
 - ▶ variable selection networks: softmax weights over the inputs at each time
 - ▶ static covariate encoders that condition every other block; an LSTM encoder–decoder for local patterns
 - ▶ interpretable multi-head attention (values shared across heads) for long-range dependence; quantile outputs trained with the pinball loss
- Its “interpretability” is the variable-selection weights and the attention pattern: useful diagnostics, not causal explanations (Section 7)

Recap: Attention and Transformers

- Attention is a data-dependent weighted average; order comes only from positional encodings; cost is quadratic in the tokens
- Zeng et al.: one linear layer beat point-token Transformers; PatchTST: patches restore the advantage; both replicate on Romanian load
- The TFT organises static, known and observed inputs with gates, selection and attention, and outputs quantiles

N-BEATS: basis expansion with residual stacking

- ▣ [Oreshkin et al. \(2020\)](#): block ℓ maps its input x_ℓ through four fully connected layers to coefficients $\theta_\ell^b, \theta_\ell^f$; backcast $\hat{x}_\ell = g^b(\theta_\ell^b)$, forecast $\hat{y}_\ell = g^f(\theta_\ell^f)$
 - ▶ doubly residual: $x_{\ell+1} = x_\ell - \hat{x}_\ell$, $\hat{y} = \sum_\ell \hat{y}_\ell$: each block explains what the previous ones left
- ▣ **Generic**: g linear and learned; **interpretable**: trend basis $\{(t/H)^j\}_{j \leq 3}$ and seasonality basis $\{\cos 2\pi jt/H, \sin 2\pi jt/H\}$: a learned exponential-smoothing-like decomposition
- ▣ Pure deep learning, no time-series features; trained globally per frequency with sMAPE/MASE losses; the published results are ensembles of 180 networks
 - ▶ their M4 result beat the competition winner, the ES-RNN hybrid of [Smyl \(2020\)](#)

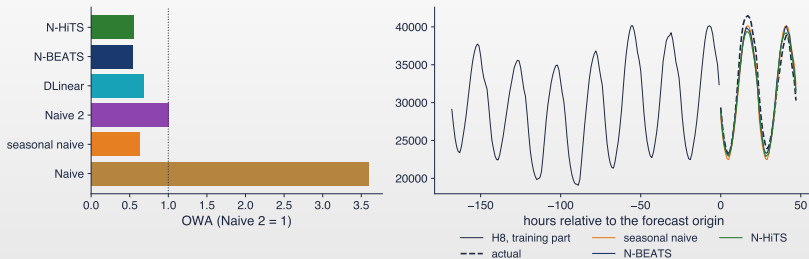
N-HiTS: multi-rate inputs, hierarchical outputs

- ▣ [Challu et al. \(2023\)](#): block ℓ first max-pools its input with kernel k_ℓ (e.g. 8, 4, 1): low-resolution blocks see the long-run shape, high-resolution blocks the detail
- ▣ Each block outputs only H/r_ℓ coefficients, interpolated to H points: the expressiveness ratio r_ℓ forces coarse blocks to forecast smooth components
 - ▶ an explicit frequency decomposition of the forecast, close in spirit to the wavelet multiresolution of [Chapter 11](#)
- ▣ Fewer parameters and less compute than N-BEATS for long horizons, with similar accuracy on short ones
- ▣ Ours: generic N-BEATS (3 blocks $\times$ 3 layers $\times$ 256 units) and N-HiTS (pools 8, 4, 1), input 336 hours, each window divided by its mean absolute level, MAE loss, median of 3 seeds

Design: the M4 hourly subset

- ▣ Makridakis, Spiliotis and Assimakopoulos (2020): 414 hourly series, horizon 48; sMAPE, MASE with seasonal period 24 Hyndman and Koehler (2006), and $OWA = \frac{1}{2} (sMAPE/sMAPE_{Naive2} + MASE/MASE_{Naive2})$
- ▣ Our Naive, seasonal naive and Naive 2 reproduce the official M4 evaluation file exactly: Naive 2 sMAPE 18.38 (official 18.38), MASE 2.40 (official 2.40)
- ▣ Reference points from the same file: the M4 MLP and RNN benchmarks (OWA 0.921 and 1.036), Theta 1.006, the winner Smyl (2020) 0.440, the runner-up Montero-Manso et al. 0.484
- ▣ All global models are trained only on the training parts: the test period is never seen

Global deep models on the M4 hourly series



□ Left: OWA (Naive 2 = 1); right: one series with the seasonal naive, N-BEATS and N-HiTS forecasts

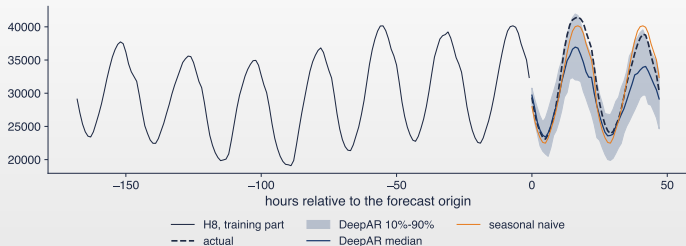
Interpreting the M4 hourly results

- OWA: seasonal naive 0.628, DLinear 0.683, N-BEATS 0.539 (sMAPE 10.73, MASE 1.19), N-HiTS 0.548
- A few minutes of CPU training put N-BEATS ahead of every M4 benchmark and of the M4 neural benchmarks (0.921, 1.036), behind the two best entries (0.440, 0.484)
- The M4 neural benchmarks were local (one network per series); ours are global: pooling 414 series is what makes the difference
- DLinear is weaker here: hourly M4 series need nonlinear interactions of level and season that one linear layer cannot express

DeepAR: probabilistic forecasts from a global RNN

- ▣ Salinas et al. (2020): $h_{i,t} = \text{LSTM}(h_{i,t-1}, z_{i,t-1}, x_{i,t})$, likelihood $z_{i,t} \sim \ell(\cdot | \theta(h_{i,t}))$: Gaussian (μ, σ) for real data, negative binomial for counts
 - ▶ training: maximise $\sum_{i,t} \log \ell(z_{i,t} | \theta(h_{i,t}))$ over random windows of all series, each scaled by $1 +$ the mean of its context
 - ▶ inputs include lagged values at seasonal lags (here 1, 24, 168) and calendar features
- ▣ Forecast: ancestral sampling, each draw fed back as the next input; quantiles from the sample paths
 - ▶ evaluated by the weighted quantile loss $\text{wQL} = \sum_{\tau} 2 \sum_{i,t} \rho_{\tau}(z - \hat{q}_{\tau}) / \sum_{i,t} |z|$, an approximation of the CRPS Gneiting and Raftery (2007)
- ▣ The model of choice for large retail and energy panels; the base of several foundation models (Chapter 13)

A small DeepAR on the M4 hourly series



- Two-layer LSTM (40 units), context 168 hours, 4 000 training steps of 64 windows, 100 sample paths; the same series as on the previous chart

Interpreting DeepAR

- wQL by training budget (steps): 500: 0.0344, 1000: 0.0338, 2000: 0.0374, 4000: 0.0405; seasonal naive with empirical quantiles: 0.0375
- After 1000 steps DeepAR beats the benchmark (80% coverage 83%); with longer training it overfits the training windows and loses (76% at 4000 steps)
- Choosing the best checkpoint on this table would be data snooping: the budget must be fixed on a validation block before the test period is opened
- Probabilistic deep models are sensitive to the training budget; report the rule that set it, and the benchmark under a proper score

Lessons from the M5 competition

- ▣ [Makridakis, Spiliotis and Assimakopoulos \(2022\)](#): 42 840 Walmart series in a hierarchy, daily unit sales, 28 days ahead; accuracy and uncertainty tracks
 - ▶ the winning accuracy methods were global LightGBM models on engineered features (lags, rolling means, prices, calendar), often combined
- ▣ Global models, cross-learning and external information (prices, events) mattered more than architecture
- ▣ The simple benchmarks were beaten, but by smaller margins than ML marketing suggests, and least at the bottom level of the hierarchy
- ▣ An earlier warning: ML methods lost to statistical ones on the M3 monthly data [Makridakis, Spiliotis and Assimakopoulos \(2018\)](#); the difference between M3 and M5 is data size and cross-series learning

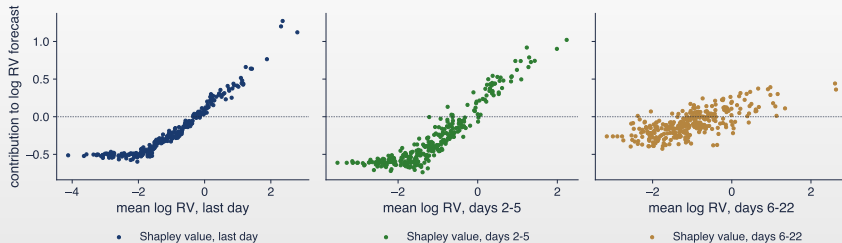
Recap: Global deep models

- ▣ N-BEATS and N-HiTS: residual stacks of fully connected blocks, generic or with trend and seasonal bases, multi-rate in N-HiTS
- ▣ On M4 hourly, global N-BEATS beats every benchmark and the local neural networks of M4
- ▣ DeepAR gives sample paths from a likelihood; it beats the seasonal naive under the wQL only at the right training budget, which must be set on validation data

Shapley values

- **Shapley (1953)**: a game $v : 2^{\{1..G\}} \rightarrow \mathbb{R}$; $\phi_g = \sum_{S \subseteq \{1..G\} \setminus g} \frac{|S|!(G - |S| - 1)!}{G!} [v(S \cup g) - v(S)]$
 - ▶ the unique allocation with efficiency ($\sum_g \phi_g = v(\text{all}) - v(\emptyset)$), symmetry, dummy and additivity
- **SHAP Lundberg and Lee (2017)**: $v(S) = \mathbb{E}[f(x_S, X_{\bar{S}})]$; **interventional**: $X_{\bar{S}}$ from a background sample, independent of x_S ; **observational**: conditional on x_S
 - ▶ with lags, features are strongly correlated: interventional values evaluate the model at impossible histories; group correlated lags instead
- Here: exact Shapley values of three lag groups (day, days 2–5, days 6–22) for boosting on S&P 500 log RV, 300 forecast days, 100 background days

What the boosted model has learned



□ Each point: one forecast day; horizontal: the mean log RV of the group; vertical: its Shapley value

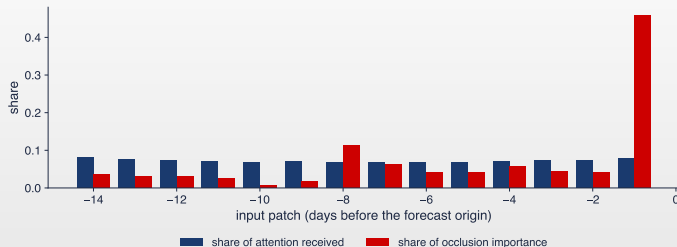
Interpreting the Shapley values

- Mean $|\phi|$ shares: last day 37%, days 2–5 46%, days 6–22 17%
- HAR on the same sample: coefficients 0.28, 0.50, 0.17 on the daily, weekly and monthly means: the boosted trees rediscovered the HAR cascade
- The values flatten at the extremes: trees cap their forecasts at the training range, the reason boosting lost to HAR in 2020
- Shapley values describe the model, not the data-generating process: a correct description of a wrong model is still wrong

Attention is not explanation

- Jain and Wallace (2019): attention weights are often uncorrelated with gradient-based importance, and very different attention patterns give the same predictions
- Wiegrefe and Pinter (2019): attention can be a faithful explanation under conditions, but this must be tested, not assumed
- A test: occlusion importance, the increase of the MSE when one input patch is replaced by the window mean, against the attention each patch receives
 - ▶ on the patch Transformer of Section 5 ($H = 96$), the first 1000 test windows

Where the Transformer looks, and what it uses



- Share of attention received by each of the 14 daily patches against the share of occlusion importance

 [ATS_ch12_interpretation](#)

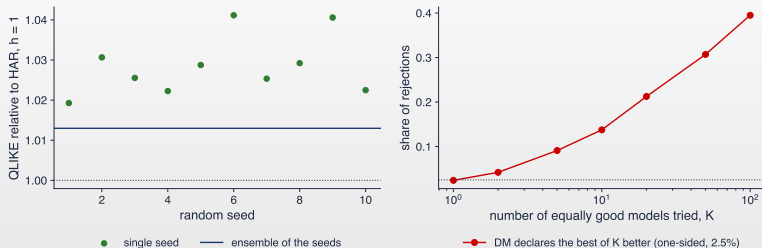
Interpreting attention and occlusion

- ▣ The last day carries 46% of the occlusion importance but receives only 8% of the attention
- ▣ Most attention goes to the patch 14 days back; the most important patch is 1 day back; Spearman correlation -0.09
- ▣ The flatten head reads the token states directly: what matters can bypass the attention weights entirely
- ▣ Report attention maps as what they are, a mixing pattern; for importance, use occlusion, Shapley or gradients, with a sanity check

The honest benchmark

- Strong baselines first: seasonal naive, ETS/Theta, HAR, an expert ARX; Lago et al. built an open benchmark because published electricity-price models were rarely compared with them [Lago et al. \(2021\)](#)
- Pitfalls catalogued by [Hewamalage, Ackermann and Bergmeir \(2023\)](#): test-set reuse, leakage in scaling and features, errors averaged across series of different scales, no significance tests, unreported tuning budget
- Tests ([Chapter 1](#)): DM for two models, MCS [Hansen, Lunde and Nason \(2011\)](#) for many, the reality check [White \(2000\)](#) and SPA [Hansen \(2005\)](#) when the best of many was chosen on the same data
- Report seeds, hardware, training time, every architecture tried; pre-register the design ([Chapter 0](#))

Seeds and data snooping



- Left: MLP against HAR for S&P 500 RV, $h=1$, with 10 seeds and their ensemble; right: simulation, K equally good models, 2500 test days, correlation 0.5

Interpreting seeds and snooping

- Seeds alone move the QLIKE ratio from 1.019 to 1.041 (median 1.027); the ensemble of the seeds reaches 1.013, better than every single seed
- Picking the best of K equally good models on the test set: the DM test declares it better (one-sided, nominal 2.5%) in 2% of cases for $K = 1$, 14% for $K = 10$, 40% for $K = 100$
- Hyper-parameter searches, seeds and architectures are all K : count them, and test the winner with SPA or MCS on data not used for the choice
- Ensembles are the cheap cure for seed variance; pre-registration is the cure for snooping

Recap: Interpretation and honest benchmarks

- ▣ Shapley values have axioms; with lagged features use groups and remember they describe the model
- ▣ Attention weights are not importance: on our Transformer they point the other way
- ▣ Strong baselines, DM/MCS/SPA, all seeds and all tries reported: otherwise any model can be made to win

An open question

- Do machine-learning and deep models forecast realised volatility better than HAR, across markets and horizons, once the comparison is pre-registered?
 - ▶ formal: $H_0: \mathbb{E}[\text{QLIKE}_{\text{ML}} - \text{QLIKE}_{\text{HAR}}] \geq 0$ in each cell; a cell supports ML only if DM rejects at 5% and the MCS excludes HAR
 - ▶ falsified if fewer than a fifth of the cells support ML
- Why it matters: risk systems (VaR, margins, option hedging) run on volatility forecasts; a spurious ML gain becomes a model risk
 - ▶ literature to start from: Corsi (2009), Bucci (2020), Christensen, Siggaard and Veliyev (2023), Hewamalage, Ackermann and Bergmeir (2023)

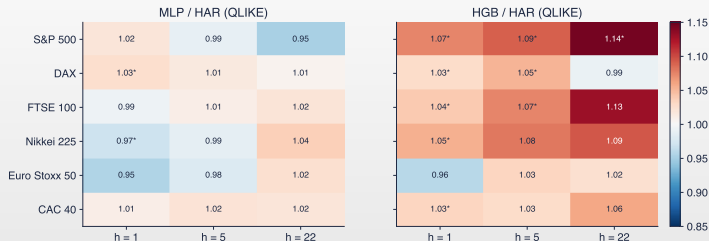
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers since 2019 comparing neural networks or boosting with HAR for realised volatility out of sample; give DOIs and the test used. Then check every DOI on Crossref
 - ▶ **design:** Write a pre-registration: assets, horizons, windows, refit schedule, loss, tests, seeds, what counts as a win.
 - ▶ **code and replication:** reproduce first the S&P 500 HAR QLIKE (0.253 at $h = 1$ here), then add learners one at a time
 - ▶ **critique:** Act as a hostile referee: list every source of leakage, snooping and unfair tuning in this comparison.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- No future information: scaling, feature construction and tuning use only data before each origin; purge around h -step targets
- The baseline is tuned as carefully as the learner, and refitted on the same schedule
- Seeds, architectures and hyper-parameters tried are all reported; the winner is tested with MCS or SPA
- An AI claim that “LSTMs capture long memory, so they beat HAR” is checked against the numbers, not accepted

Mini-case: do learners beat HAR across markets?



- 36 cells: six indices $\times$ three horizons $\times$ two learners (MLP, boosting), the expanding design of Section 4; QLIKE ratio to HAR, a star where DM rejects at 5%
- Ratios from 0.95 to 1.14, median 1.024 (MLP 1.009, boosting 1.052); 9 cells below 1, 1 significantly better, 10 significantly worse: the hypothesis is falsified for these learners

Project idea

- ▣ **Deep learning against HAR for Central and Eastern European volatility:** replicate first, then extend
 - ▶ replicate: [Corsi \(2009\)](#) HAR and the neural comparison of [Bucci \(2020\)](#) on the Oxford-Man indices, with the design of Section 4
 - ▶ extend: intraday-based RV for the BET, WIG20 and EUR/RON; global models across markets; exogenous predictors (VIX, macro news) as in [Christensen, Siggaard and Veliyev \(2023\)](#)
 - ▶ pre-register: markets, horizons, windows, every learner and its tuning budget, QLIKE, DM, MCS, seeds
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Validation must respect dependence: K-fold only for well-specified autoregressions; otherwise block, purge and keep a final test period
- ▣ Global models pool series and afford memory; high-dimensional selection on short macro samples is unstable
- ▣ Trees and networks rarely beat strong structured baselines alone (expert ARX, HAR); they help in combinations and in large panels
- ▣ Architecture details matter: open forget gates, patches instead of point tokens, global training
- ▣ Attention is not explanation; seeds and searches are hidden multiple testing

Self-assessment

Questions

- Why is random K-fold valid for an AR(3) fitted to AR(3) data but not for 22-day targets?
- Which condition does the adaptive lasso remove, and how?
- Why does adding trees not reduce the variance of a forest below $\rho\sigma^2$?
- What is the receptive field of a TCN with $k = 3$ and five levels?
- Why can a patch Transformer beat DLinear when a point-token Transformer cannot?

Next:

- Foundation models and conformal prediction
- pretrained models used zero-shot, and distribution-free intervals for any forecaster, including the ones of this chapter

Appendix: the bias of K-fold under autocorrelated errors

- Linear model $y = X\beta + \varepsilon$; leave out row i : $e_{(i)} = e_i / (1 - h_{ii})$, h_{ii} the leverage
- $\mathbb{E} e_{(i)}^2 \approx \sigma^2(1 + h_{ii}) - 2 \sum_{j \neq i} \frac{H_{ij}}{1 - h_{ii}} \text{Cov}(\varepsilon_i, \varepsilon_j)$ to first order
- Uncorrelated errors: the second term vanishes and LOOCV estimates the out-of-sample MSE $\sigma^2(1 + h)$: the Bergmeir–Hyndman–Koo case
- Positive correlation between the left-out error and the neighbouring training errors ($H_{ij} > 0$ for neighbours with similar lags): the estimate is too small; purging sets the neighbouring terms to zero

Appendix: soft thresholding and the elastic net

- ▣ With $X'X = nI$ the objective separates: $\frac{1}{2}(\beta_j - z_j)^2 + \lambda|\beta_j|$ for each j , $z_j = x_j'y/n$
- ▣ Subgradient: $0 \in \beta_j - z_j + \lambda \partial|\beta_j|$; if $|z_j| \leq \lambda$, $\beta_j = 0$; otherwise $\beta_j = z_j - \lambda \text{sign}(z_j)$
- ▣ Elastic net: $\frac{1}{2}(\beta_j - z_j)^2 + \lambda\alpha|\beta_j| + \frac{\lambda(1-\alpha)}{2}\beta_j^2$ gives $\beta_j = S(z_j, \lambda\alpha)/(1 + \lambda(1 - \alpha))$
- ▣ Adaptive lasso: threshold $\lambda w_j = \lambda/|\tilde{\beta}_j|^\gamma$, small for large $|\tilde{\beta}_j|$: the bias on large coefficients vanishes, which gives the oracle property

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